WORKED SOLUTION

# STEP I Specimen paper, Question 16

Probability and Statistics • Continuous Random Variables

**STEP I Specimen paper, Question 16**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q16

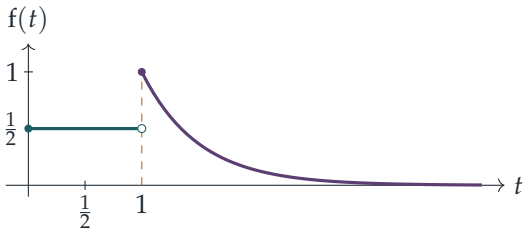
## The key idea

The unknown $k$ comes from the fact that the total probability is 1, and everything else follows from integrating the two pieces of the density separately. For the cost, the trick is to turn the continuous variable $T$ into a discrete one: the number of coins depends only on how many complete half-minutes have passed, so we need the probability that $T$ lands in each half-minute interval. A geometric-type series then does the rest.

## Solution

### Finding $k$ and the mean

The density is a flat piece of height $\frac{1}{2}$ on $[0,1)$ followed by an exponential tail, with a jump at $t = 1$.




The first piece contributes probability $\frac{1}{2}$, so the tail must contribute the other $\frac{1}{2}$:

$$k\int_1^\infty \mathrm{e}^{-2t}\,\mathrm{d}t = k\left[-\tfrac{1}{2}\mathrm{e}^{-2t}\right]_1^\infty = \tfrac{1}{2}k\mathrm{e}^{-2} = \tfrac{1}{2}, \qquad \text{so} \qquad k = \mathrm{e}^2.$$

(The density on $t \geq 1$ is therefore $\mathrm{e}^{2-2t}$, which equals 1 at $t = 1$, as in the sketch.)

For the mean, integrating by parts with $u = t$, $\mathrm{d}v = \mathrm{e}^{-2t}\,\mathrm{d}t$:

$$\int_1^\infty t\,\mathrm{e}^{-2t}\,\mathrm{d}t = \left[-\tfrac{1}{2}t\,\mathrm{e}^{-2t}\right]_1^\infty + \tfrac{1}{2}\int_1^\infty \mathrm{e}^{-2t}\,\mathrm{d}t = \tfrac{1}{2}\mathrm{e}^{-2} + \tfrac{1}{4}\mathrm{e}^{-2} = \tfrac{3}{4}\mathrm{e}^{-2},$$

using $t\,\mathrm{e}^{-2t} \to 0$ as $t \to \infty$. Hence

$$\mathrm{E}(T) = \int_0^1 \tfrac{1}{2}t\,\mathrm{d}t + \mathrm{e}^2 \cdot \tfrac{3}{4}\mathrm{e}^{-2} = \tfrac{1}{4} + \tfrac{3}{4} = 1,$$

so the expected length of a call is one minute, as required.

#### The cumulative distribution function

Write $\mathrm{F}(t) = \mathrm{P}(T \le t)$. For $t < 0$ it is 0; for $0 \le t < 1$ it is $\int_0^t \tfrac{1}{2}\,\mathrm{d}s = \tfrac{1}{2}t$; and for $t \ge 1$,

$$\mathrm{F}(t) = \tfrac{1}{2} + \int_1^t \mathrm{e}^{2-2s}\,\mathrm{d}s = \tfrac{1}{2} + \tfrac{1}{2}(1 - \mathrm{e}^{2-2t}) = 1 - \tfrac{1}{2}\mathrm{e}^{2-2t}.$$

So

$$\mathrm{F}(t) = \begin{cases} 0 & t < 0, \\ \tfrac{1}{2}t & 0 \le t < 1, \\ 1 - \tfrac{1}{2}\mathrm{e}^{2-2t} & t \ge 1. \end{cases}$$

Note that F is continuous at $t = 1$ (both formulas give $\tfrac{1}{2}$) even though f jumps there, and $\mathrm{F}(t) \to 1$ as $t \to \infty$. A form we will use: for $t \ge 1$, $\mathrm{P}(T > t) = \tfrac{1}{2}\mathrm{e}^{2-2t}$.

#### The expected cost

One coin goes in at the start, and one more each time a half-minute is completed. So if the call lasts $T$ minutes, the number of complete half-minutes is $\lfloor 2T \rfloor$ and

$$C = 10(1 + \lfloor 2T \rfloor).$$

(Whether a coin is inserted at the exact instant the call ends at a half-minute mark doesn't matter: $T$ is continuous, so that has probability 0.)

**Ask yourself** How do we get from a continuous $T$ to the distribution of $C$?

The event $C = 10(n+1)$ is exactly the event $\tfrac{n}{2} \le T < \tfrac{n+1}{2}$, and we can read its probability off F:

- $n = 0$: $\mathrm{P}(C = 10) = \mathrm{F}(\tfrac{1}{2}) = \tfrac{1}{4}$;
- $n = 1$: $\mathrm{P}(C = 20) = \mathrm{F}(1) - \mathrm{F}(\tfrac{1}{2}) = \tfrac{1}{4}$;
- $n \ge 2$: both endpoints are at least 1, so

$$\mathrm{P}\big(C = 10(n+1)\big) = \tfrac{1}{2}\mathrm{e}^{2-n} - \tfrac{1}{2}\mathrm{e}^{2-(n+1)} = \tfrac{1}{2}\big(1 - \mathrm{e}^{-1}\big)\mathrm{e}^{2-n}.$$

Put $r = \mathrm{e}^{-1}$ and $m = n - 2$, so that $\mathrm{e}^{2-n} = r^m$ and $10(n+1) = 10(m+3)$. Then

$$\begin{aligned} \mathrm{E}(C) &= 10 \cdot \tfrac{1}{4} + 20 \cdot \tfrac{1}{4} + \sum_{m=0}^{\infty} 10(m+3) \cdot \tfrac{1}{2}(1-r)\,r^m \\ &= \tfrac{15}{2} + 5(1-r)\left(\sum_{m=0}^{\infty} m r^m + 3\sum_{m=0}^{\infty} r^m\right). \end{aligned}$$

Since $0 < r < 1$, the geometric series gives $\sum r^m = \dfrac{1}{1-r}$, and differentiating it term by term (then multiplying by $r$) gives $\sum m r^m = \dfrac{r}{(1-r)^2}$. Therefore

$$\mathrm{E}(C) = \tfrac{15}{2} + 5(1-r)\left(\frac{r}{(1-r)^2} + \frac{3}{1-r}\right) = \tfrac{15}{2} + \frac{5r}{1-r} + 15.$$

Finally $\dfrac{r}{1-r} = \dfrac{\mathrm{e}^{-1}}{1-\mathrm{e}^{-1}} = \dfrac{1}{\mathrm{e}-1}$, so

$$\mathrm{E}(C) = 22\tfrac{1}{2} + \frac{5}{\mathrm{e}-1},$$

as required. Numerically this is about 25.41 pence.

#### The comment

At 20 pence per minute, a call of mean length one minute would cost 20 pence on average, but $\mathrm{E}(C) \approx 25.4$. The difference arises because the man pays *in advance* for each half-minute: every coin buys a full half-minute, but the last one is almost never used up. Indeed $C = 10(1 + \lfloor 2T \rfloor) > 20T$ for every call, since $\lfloor 2T \rfloor > 2T - 1$; the cost rounds the length *up* to the next half-minute, so $\mathrm{E}(C)$ exceeds $20\,\mathrm{E}(T)$ by twenty times the expected unused paid time (about 5.4 pence here).

## Checks

- The probabilities of $C$ add to 1: $\frac{1}{4} + \frac{1}{4} + \sum_{m \geq 0} \frac{1}{2}(1-r)r^m = \frac{1}{2} + \frac{1}{2} = 1$.
- The bound $20T < C \leq 20T + 10$ gives $20 < \mathrm{E}(C) \leq 30$, and 25.41 sits comfortably inside. If the unused time in the last half-minute were uniform, we would expect an excess of about 5 pence; the long tail gives slightly more.
- Numerically, Simpson's rule gives total probability 1 and mean 1, and a simulation of $10^6$ calls (half uniform on $[0, 1)$, half 1 plus an exponential with rate 2) gave average cost 25.41 pence, matching $22.5 + 5/(\mathrm{e}-1) = 25.4099$.

## Common slips

- Taking the number of coins to be $\lfloor 2T \rfloor$ rather than $1 + \lfloor 2T \rfloor$, forgetting the coin at the start; this loses exactly 10 pence from the answer.
- Using the $t \geq 1$ formula for F at $t = \frac{1}{2}$: the intervals $[0, \frac{1}{2})$ and $[\frac{1}{2}, 1)$ must be handled with the uniform piece.

## Beyond STEP

Notice that the tail of $T$ is memoryless: given $T \geq 1$, the remaining time $T - 1$ is exponential with rate 2, which is why the probabilities for $n \geq 2$ form a geometric

sequence. Writing $N = \lfloor 2T \rfloor$, the tail-sum formula $\mathrm{E}(N) = \sum_{n \geq 1} \mathrm{P}(N \geq n)$ gives the same answer even faster. Mixtures of discrete and continuous behaviour like this are treated carefully in Grimmett and Welsh, *Probability: An Introduction*, and Blitzstein and Hwang, *Introduction to Probability*, has many more examples of the memoryless property.

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